PHYSICS

SECTION – I
Single Correct Choice Type

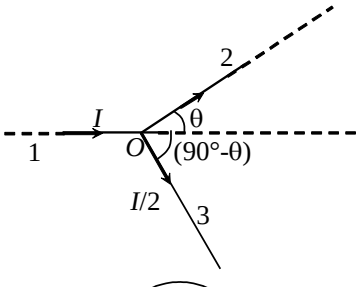
This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. Find the value of θ in the range of $0 \le \theta \le \frac{\pi}{2}$, so that the magnetic field is maximum at point *P* (0, 0, *d*)

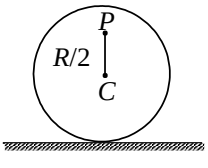
(a) 45° (b) 60°

(c) 30° (d) 90°



2. A cylinder of mass *M* and radius *R* is rolling over a smooth horizontal surface. A point *P* is on the cylinder a distance $\frac{R}{2}$ above the center of the cylinder. The angular momentum of the cylinder about point *P* is (the velocity of its CM is v_0)

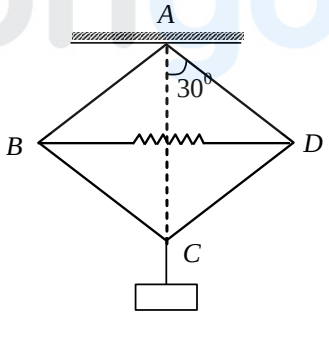
(a) zero (b) $\frac{MRv_0}{4}$ (c) MRv_0 (d) $\frac{3}{4}mRv_0$



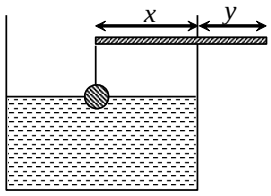
3. Four weightless rods of length *l* each are connected by hinged joints and forms a rhombus. Hinge *A* is fixed and load is suspended from hinge *C*. Hinge D and B are connected by weightless spring of length 1.5 *l* in unstretched state. In equilibrium the rod forms an angle 30^0 with vertical, then the time period of oscillation of load is

(a) $T = 2\pi\sqrt{\frac{l\sqrt{3}}{g}}$ (b) $T = 2\pi\sqrt{\frac{3\sqrt{3}l}{2g}}$

(c) $T = 2\pi\sqrt{\frac{l}{2\sqrt{3}g}}$ (d) $T = 2\pi\sqrt{\frac{l\sqrt{3}}{5g}}$



4. A homogenous aluminium ball of radius r = 0.5cm is suspended by a weightless thread from an end of a homogenous rod of mass M = 4.4 gm. The rod is placed on the edge of a tumbler with water so that half of the ball is submerged in water when the system is in equilibrium (figure shown). Densities of aluminium and water is 2.7 × 10^3 kg /m^3 and 10^3 kg/m^3 respectively. The ratio $\frac{y}{x}$ of the parts of the brim is approximately



(a) $\frac{3}{2}$ (b) $\frac{2}{3}$ (c) $\frac{10}{27}$ (d) $\frac{27}{10}$

5. Temperature dependence of speed of sound could be expressed as $v = k\sqrt{T}$, where k is some positive constant. In a medium sound is traveling along x-axis. At x=0 temperature is T_1 and at $x = l$, temperature is T_2 and the temperature varies linearly with x. Time taken by sound to travel from $x = 0$ to $x = l$ is

(a) $\frac{2l}{k\left(\sqrt{T_1}+\sqrt{T_2}\right)}$ (b) $\frac{l}{k\left(\sqrt{T_1}+\sqrt{T_2}\right)}$ (c) $\frac{l}{2k\left(\sqrt{T_1}+\sqrt{T_2}\right)}$ (d) $\frac{1}{2k\left(\sqrt{T_1}+\sqrt{T_2}\right)}$

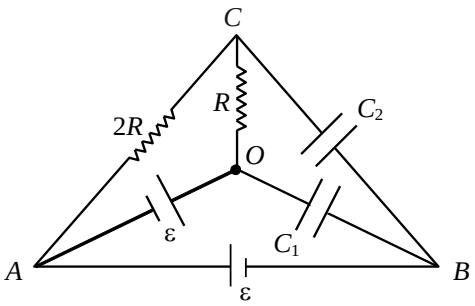
6. Molar heat capacity of gas whose molar heat capacity at constant volume is C_V for process $P = 2e^{2V}$ is

(a) $C_V + \frac{R}{1+2V}$ (b) $C_V + \frac{R}{2V}$ (c) $C_V + \frac{R}{V}$ (d) none of these

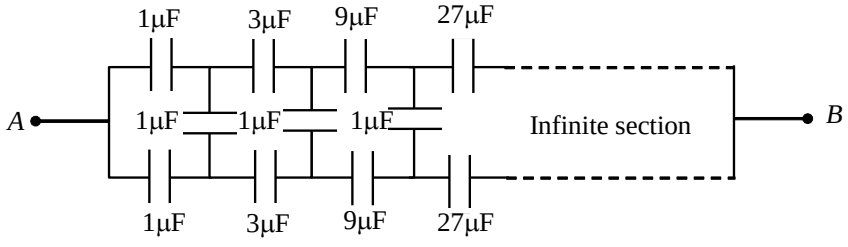
7. If energy stored in the capacitor C_1 and C_2 are same then what is the value of C_1 / C_2 ?

(a) $\frac{25}{36}$ (b) $\frac{1}{36}$

(c) $\frac{4}{9}$ (d) $\frac{9}{4}$



8. What is the equivalent capacitance between the points *A* and *B*?



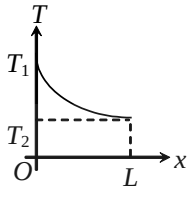
(a) $\frac{2}{3}\mu F$ (b) $\frac{4}{3}\mu F$ (c) infinite (d) $\left(1+\sqrt{3}\right)\mu F$

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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9. The two ends of a long bar maintained at different temperature. The graph of temperature against distance of the bar when it has attained steady state is shown in the figure. Then select the correct alternative(s)



(a) the cross-section area of the bar may increase as the distance from the hot end increases
(b) the cross-section area of the bar may decreases as the distance from the hot end increases
(c) the conductivity of the rod may be increasing as the distance from hot end increases
(d) the conductivity of the bar may decrease as the distance from hot end increases

10. Two flat worms *A* and *B* each of mass 20 grams climb over a very thin wall of height 10 cm. The worm *A* is 20 cm long and the worm *B* is wider and only 10 cm long. Work done by gravity on the worm *A* is W_1 when half of its is over the top of the wall and W_2 be work done by the worm *B* when half of it is over the top of the wall, then ($g = 10\ m/s^2$)

(a) $W_1 = -10.0$ mJ (b) $W_2 = -15.0$ mJ
(c) $W_2 = 15.0$ mJ (d) $W_1 = 20.0$ mJ

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11. The threshold wavelength for photoelectric emission from a material is 6000 A°. Photo electrons will be emitted when this material is illuminated with a laser beam of
(a) red colour and having power 50 W
(b) blue colour and having power 50 W
(c) blue colour and having power 0.005 W
(d) violet colour and having power 5 W

12. A diatomic gas goes through a process $\Delta U + \lambda W = 0$. Now if
(a) $\lambda = 1$, then process is adiabatic
(b) $\lambda = 0$, then process is isochoric
(c) $\lambda = -\frac{5}{2}$, then the process is isobaric
(d) $\lambda = \frac{5}{2}$, then the process follows $PV^2 =$ constant

SECTION–III
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

A particle *P*, of charge *q* and mass *m*, is placed at a point on a smooth horizontal surface and is free to move. Another particle *Q*, of same charge and mass 2*m* is projected from a distance *r* from *P* with an initial speed v_0 directly towards *P*. As *Q* approach *P*, first the separation between them decreases and after that increases.

13. What is the speed of particle *P* when separation between them is minimum?
(a) zero (b) $\frac{v_0}{2}$ (c) $\frac{v_0}{4}$ (d) $\frac{2v_0}{3}$

14. At the closest separation, the potential energy of the system of particles is (Here r_0 is minimum separation between the particles)

(a) $\frac{q^2}{4\pi \in_0 r}$ (b) $\frac{q^2}{4\pi \in_0 r}+\frac{mv_0^2}{3}$

(c) $\frac{-q^2}{4\pi \in_0 r_0}+\frac{1}{4}mv_0^2$ (d) $\frac{q^2}{4\pi \in_0 r}+\frac{1}{2}mv_0^2$

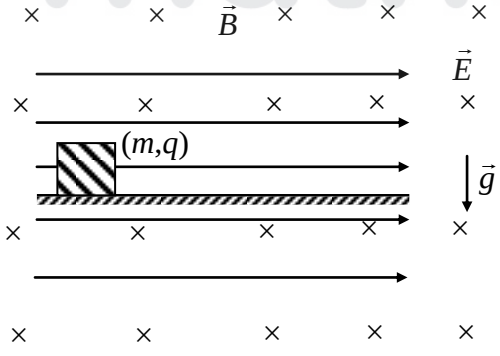
15. Separation between the two particles when speed of P is $\frac{v_0}{2}$

(a) $\frac{q^2}{\frac{q^2}{r}+\frac{5mv_0^2\pi \in_0}{4}}$ (b) $\frac{1}{\frac{1}{r}+\frac{3mv_0^2\pi \in_0}{4q^2}}$

(c) $\frac{16qr}{4q^2+5mv_0^2\pi \in_0 r}$ (d) $\frac{r}{10}$

Passage-II

A small insulating block having mass m = 100 gram carrying charge q = +1mC is placed at rest on a smooth insulating horizontal surface in uniform vertical gravitational field (g = 10 ms^{-2}). At t = 0, uniform and constant electric and magnetic field having magnitudes $E = 10^4$ N/C and B = 0.1 Tesla is switched on in the directions as shown and the block is free to move. For subsequent motion of the block in the given situation answer following questions.



16. The block will lose contact with the horizontal surface at

(a) $t = 1$ s (b) $t = 10$ s (c) $t = 100$ s (d) $t = 1000$ s

17. The block will lose contact with the surface after travelling through

(a) 5m (b) 5×10^2m (c) 5×10^4m (d) 5×10^5m

18. Magnitude of time rate of change of normal reaction on the block by the surface will be

(a) 10^{-2} N/s (b) 5×10^{-3} N/s (c) 10^{-3} N/s (d) 0.1 N/s

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

	1	2	3	4	5
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

1. Column–I contains equation of a particle representing different motions. Here $\vec{v}$ represents instantaneous velocity x is position along x axis, y is position along y axis and t is an instant of time. Column–II contains various characteristics of the motion. Match appropriately and mark your answer

Column–I		Column–II	
(A)	$\vec{v} = 5t\,\hat{i} + 4\,t\,\hat{j}$	**1.**	Straight line motion
(B)	$\vec{v} = -5t\,\hat{i} + 5\,\hat{j}$	**2.**	Motion on parabolic path
(C)	$y = 2x - 5x^2$	**3.**	Acceleration is constant
(D)	$x = 6\sin\frac{2\pi t}{5},\ y = 6\cos\frac{2\pi t}{5}$	**4.**	Velocity is not constant
		5.	Speed is constant

2. When we write expression for energy of electron in n^{th} orbit of helium ion (He^+) we take zero potential energy for $n = \infty$, but the potential energy depends on reference. If we take total energy of atom for $n = 1$ orbit as zero then

Column–I		Column–II	
(A)	Total energy of electron in $n = 2$	**1.**	54.4 eV
(B)	Ionization energy from ground state	**2.**	40.8 eV
(C)	Energy required to excite electron from $n = 1$ to $n = 2$	**3.**	depends on reference level
(D)	Negative of potential energy of electron in $n = 1$	**4.**	independent of reference level.
		5.	65.28×10^{-20} J

CHEMISTRY

SECTION – I

Single Correct Choice Type

This section contains 4 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. A sample of sea water containing 2.8% NaCl by mass has density 1.03 g/cm^3. A saturated solution of NaCl in water is 5.45 M. How much water would have to be evaporated from 4 L of sea water before NaCl would start precipitating?

(a) 3.64 L (b) 0.36 L (c) 2.64 L (d) 1.36 L

2. A silver electrode is immersed in a saturated $Ag_2SO_4(aq)$ solution. The potential difference between silver electrode and standard hydrogen electrode is found to be 0.74 V. The K_{sp} of Ag_2SO_4 would be (Given that $E^{o}_{Ag^+/Ag} = 0.799$ V)

(a) 5×10^{-6} (b) 5×10^{-5} (c) 5×10^{-4} (d) 5×10^{-3}

3. $CH_3-\overset{O}{\overset{\|}{C}}-Cl + CH_3-CH_2-\underset{CF_3}{\underset{|}{CH}}-COOEt \xrightarrow[(ii)\ H^+/H_2O]{(i)\ OH^-} (A) \xrightarrow{\Delta} (B) \xrightarrow{C_6H_5-\overset{O}{\overset{\|}{C}}-\overset{18}{O}-O-H} (C)$

Product (C) would be

(a) $CH_3-\overset{O^{18}}{\overset{\|}{C}}-O-\underset{CF_3}{\underset{|}{CH}}-C_2H_5$ (b) $CH_3-O-\overset{O^{18}}{\overset{\|}{C}}-\underset{CF_3}{\underset{|}{CH}}-C_2H_5$

(c) $CH_3-\overset{O}{\overset{\|}{C}}-\overset{18}{O}-\underset{CF_3}{\underset{|}{CH}}-C_2H_5$ (d) $CH_3-\overset{18}{O}-\overset{O}{\overset{\|}{C}}-\underset{CF_3}{\underset{|}{CH}}-C_2H_5$

4. Naturally occuring potassium contains 1.2×10^{-2} % of ^{40}K. The human body contains 3.5×10^{-1} % of potassium by weight. The total radioactivity resulting from ^{40}K decay in 80 kg human would be ($t_{1/2}$ for $^{40}K = 6.9 \times 10^{14}$ min^{-1})

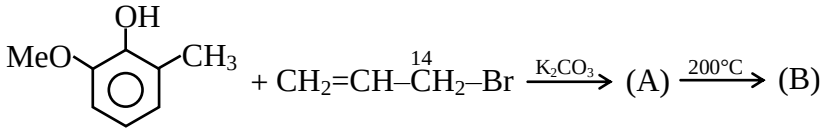
(a) 5.06×10^3 dpm (b) 5.06×10^4 dpm
(c) 5.06×10^5 dpm (d) 5.06×10^6 dpm

SECTION – II
Multiple Correct Choice Type

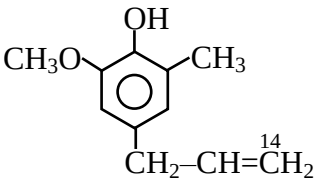
This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

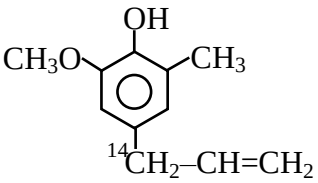
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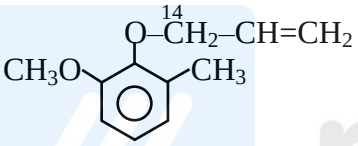
5. Product (B) of the following reaction

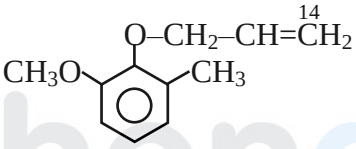


would be

(a) 

(b) 

(c) 

(d) 

6. An inorganic compound (A) is formed by passing a gas (B) through a concentrated liquor containing sodium sulphide and sodium sulphite. On adding (A) into dilute solution of $AgNO_3$, a white precipitate appears which quickly changes into black coloured compound (C). Which of the following is/are correct statement(s)?

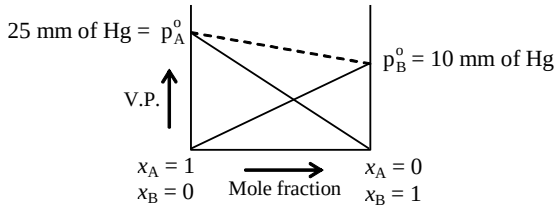
(a) (B) is NH_3 and (C) is $(NH_4)_2S$.

(b) (A) is $Na_2S_2O_3$ and white precipitate is of $Ag_2S_2O_3$.

(c) (B) is I_2 and (C) is Ag_2S.

(d) (A) is Na_2SO_4 and white precipitate is of Ag_2SO_4.

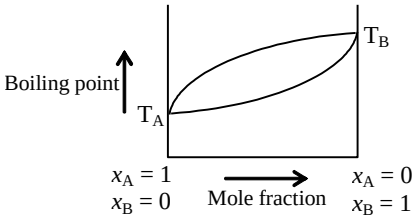
7. Two liquid (A) and (B) when mixed in any proportion forms an ideal solution. A graph vapour pressure vs mole fraction for this solution is shown as under:



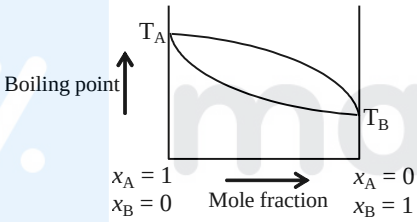
Which of the following statement(s) is/are correct?

(a) From above graph, it is evident that mole fraction (A) in vapour state is always higher than (B) irrespective of composition of solution.

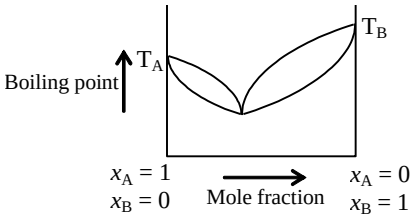
(b) Graph of boiling point vs mole fraction for this solution would be



(c) Graph of boiling point vs mole fraction for this solution would be:



(d) If (A) and (B) solution have shown positive deviation then their boiling point vs mole fraction graph would have been:



8. In which of the following reaction N_2 gas is produced.

(a) $NH_4Cl + NaNO_2 \xrightarrow{\Delta}$ (b) $CaOCl_2 + NH_3 \xrightarrow{\Delta}$

(c) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$ (d) $CaCN_2 + H_2O \xrightarrow{\Delta}$

SECTION – III
Reasoning Type

This section contains 4 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True ; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True ; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

9. **Statement-1:** Hydrate of cyclopropanone is readily isolated from its aqueous solution.

Statement-2: Aldehydes and ketones, undergoes reversible hydration in aqueous solution.

(a) A (b) B (c) C (d) D

10. **Statement-1:** Van't Hoff factor (i), for Mercury (I) nitrate on its complete dissociation would be three.

Statement-2: Mercurous ion exist as a dinuclear metal ion.

(a) A (b) B (c) C (d) D

11. **Statement-1:** When electrolysis of H_2O and D_2O mixture is done, H_2 is liberated much faster than D_2 and the remaining water thus becomes enriched in D_2O.

Statement-2: Natural abundance of ^1_1H is nearly 99.9 times the ^2_1D.

(a) A (b) B (c) C (d) D

12. **Statement-1:** Barium salts are toxic to us, still examination of digestive tract by doctors are done by giving a 'cocktail' containing $BaSO_4$.

Statement-2: $BaSO_4$ is opaque to X-ray. So, digestive tract is easily examined by X-ray.

(a) A (b) B (c) C (d) D

SECTION–IV
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

0.369 g of a bromoderivative of a saturation hydrocarbon (A) when vaporized occupied 67.2 mL at STP. (A) on reaction with aqeuous NaOH gives compound (B). (B) when passed over alumina at 250°C gives a neutral compound (C) while at 350°C it gives hydrocarbon (D). (D) when heated with HBr gives an isomer of (A), which on treatment with alcoholic KCN followed by acidic hydrolysis gives (E). Treatment of (E) with one equivalent of Br_2/Red P (HVZ reaction) gives α-bromo acid which is optically inactive.

13. What is the molecular formula of (A)?

(a) C_3H_7Br (b) C_4H_9Br

(c) C_2H_5Br (d) $C_5H_{11}Br$

14. The compound (E) is

(a) $CH_3CH_2CH_2COOH$ (b) $(CH_3)_2CHCH_2CH_2COOH$

(c) CH_3CH_2COOH (d) $(CH_3)_2CHCOOH$

15. Compound (C) on reaction with one equivalent of HBr gives

(a) (A) only (b) Both (A) and (D)

(c) Both (B) and (D) (d) Both (A) and (B)

Passage-II

Some of the important applications of gases are dissussed in the following paragraph.

1. Most automobiles are now equipped with air bags to protect the driver and passangers in the event of a head-on or side crash. Such bags are inflated with nitrogen gas, which is generated by the explosive decomposition of sodium azide.

   $2NaN_3(s) \longrightarrow 2Na(s) + 3N_2(g)$

   The air bag is fully inflated in about 0.05 second. This is important because the typical automobile collision lasts about 0.125 seconds.
2. A hot air ballon rises because the heated air has a lower density than surroudning air, which result in upward movement of balloon. This form of transportation is now a days is used for meterological research. Density of gas is found as $d = \frac{m}{V} = \frac{PM}{RT}$.

   The siginificance of gas density has been revealed in several tragic events. Once in African country of Cameroon in 1984 when take Nyos expelled a huge bubble of CO_2 into the atomosphere. Because CO_2 is denser than air, the CO_2 cloud hugged the ground, killing 1700 people nearby.

16. You are asked to design an air bag for a car. You know that the bag should be filled with gas at a pressure higher than atmospheric pressure, say 829 mm Hg, at a temperature of 22°C. The bag has a volume of 45.5 L. What quantity of sodium azide should be used to generate the required quantity of gas?

(a) 22.2 g (b) 44.4 g (c) 88.8 g (d) 66.6 g

17. Helium filled balloons are used to carry scientific instruments high into the atmosphere. Suppose a balloon is launched when the temperature is 22.5°C and the barometric pressure is 754 mm of Hg. If the balloon's volume is 4.19×10^3 L (and no He escapes from balloon), then what will the volume of balloon at the height of 20 km where pressure is 76 mm of Hg and temperature is –33°C?

(a) 3.38×10^6 L (b) 3.82×10^2 L (c) 3.12×10^4 L (d) 3.65×10^5 L

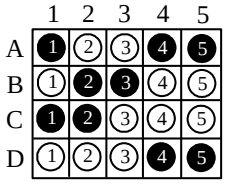
18. Which of the following interpretation(s) or observation(s) are correct for extinguishing fire by CO_2 gas?

(a) CO_2 is denser than air that is why it settles on top of air and smothers it.
(b) When CO_2 released from tank, it expands and cools significantly.
(c) White cloud seen is a condensed moisture from the air, when CO_2 is released.
(d) All are correct.

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

Column I	Column II
(A) Sol of $Fe(OH)_3$ containing $FeCl_3$	**1.** Positive colloid.
(B) Sol of AgI containing KI	**2.** Negative colloid.
(C) Colloidal sol of gold metal	**3.** Formed by peptization.
(D) Sol of As_2S_3 containing H_2S	**4.** Formed by chemical method.
	5. Formed by Bredig arc method.

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2.

Column I	Column II
(A) ΔS (entropy change)	**1.** State function
(B) ΔH (enthalpy change)	**2.** Extensive property
(C) T (temperature)	**3.** $C_p dT$
(D) V (volume)	**4.** $\left(\frac{\partial H}{\partial S}\right)_P$
	5. Intensive property

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MATHEMATICS

SECTION – I
Single Correct Choice Type

This section contains 6 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. A bag contains 10 white and 10 black balls. A person draws two balls at a time without replacement and repeats the process till bag is empty. The probability that he draws the balls of same colour each time is

(a) $\left(\frac{2}{3}\right)^{10}$ (b) $\frac{^{10}C_5}{^{20}C_{10}}$

(c) $\left(\frac{2^{10}-1}{^{20}C_{10}}\right)$ (d) none of these

2. If $f(x)=\frac{\{x\}\ g(x)}{\{x\}\ g(x)}$ is a periodic function with period $\frac{1}{4}$, where $g(x)$ is a differentiable function, then (where {.} represents fractional part of x)

(a) $g'(x)=0$ has exactly three roots in $\left(\frac{1}{4},\frac{5}{4}\right)$

(b) $g(x)=0$ at $x=\frac{k}{4}$, where $k\in I$

(c) $g(x)$ must be non-zero function

(d) $g(x)$ must be periodic function

3. If $f'\left(\frac{x}{y}\right).f\left(\frac{y}{x}\right)=\frac{x^2+y^2}{xy}\ \forall\ x,y\in R^+$ and $f(1)=1$, then $f^2(x)$ is

(a) $x+\ln x$ (b) $x^2+2\ln x$

(c) $x^2+\ln x$ (d) $x+2\ln x$

4. If $(x^2+2x+4)^n=\sum_{r=0}^{2n}a_r x^r$, then $\sum_{r=n}^{2n}\left(\frac{a_{2n-r}}{a_r}\right)$ is

(a) $2^{2n+1}-1$ (b) $\frac{4^{n+1}-1}{3}$

(c) 4^{n-1} (d) $\frac{4^{n-1}+1}{2}$

5. The straight line $\frac{x}{4}+\frac{y}{3}=1$ intersects the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ at two points *A* and *B*, there is a point *P* on this ellipse such that the area of ΔPAB is equal to $6(\sqrt{2}-1)$, the number of such points (*P*) is

(a) 1 (b) 2

(c) 3 (d) 4

6. The number of ordered pair (*a*, *b*) where $a, b \in R$ such that $(a+ib)^{2010}=a-ib$ is

(a) 2010 (b) 2011

(c) 2009 (d) 2012

SECTION – II
Multiple Correct Choice Type

This section contains 5 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

7. Let $f(x)=|x|$, $g(n,x)=\sin\left(\pi\left([n]+[n]^2\right)^{1/x}\right)$ $h(x)=\frac{\ln\left(\frac{e^x+e^{2x}+e^{3x}}{3}\right)}{x}$, then

(where [.] represent greatest integer function)

(a) $\lim_{n\to\infty}\lim_{x\to 0} f(g(n,h(x)))=1$ (b) $\lim_{n\to\infty}\lim_{x\to 0} f(g(n,h(x)))$ does not exist

(c) $\lim_{n\to\infty}\lim_{x\to 0} g(n,h(x))$ does not exist (d) $\lim_{n\to\infty}\lim_{x\to 0} g(n,h(x))=1$

8. The curve $y=(|x|-1)\operatorname{sgn}(x-1)$ divides $\frac{9x^2}{64}+\frac{4}{25}y^2=\frac{1}{\pi}$ in two parts having area A_1 and A_2 (where $A_1 < A_2$), then

(a) $\frac{A_1}{A_2}=\frac{7}{13}$ (b) $\frac{A_1}{A_2}=\frac{3}{7}$

(c) $A_1=\frac{7}{3}$ (d) $A_2=\frac{13}{7}$

9. Let $f(n)$ be the sum of first *n* terms of sequence 0, 1, 1, 2, 2, 3, 3, 4, 4 ……., then

(a) $f(n)=\frac{n^2}{4}$, where *n* is an even number

(b) $f(n)=\frac{n^2-1}{4}$, where *n* is an odd number

(c) $f(n+m)-f(n-m)=nm$, where $n, m \in I^+ (n > m)$

(d) $f(n+m)-f(n-m)=\frac{4nm+1}{1}$, where $n, m \in I^+ (n > m)$

10. Let $f_1(x)=\sin^{-1}\left(\cos(\sin^2 x)\right)$, $f_2(x)=\cos^{-1}(\sin(\cos^2 x))$
$f_3(x)=\sin^{-1}\left(\cos(\cos^2 x)\right)$, $f_4(x)=\cos^{-1}(\sin(\sin^2 x))$, then choose the correct option(s)

(a) $\sum_{i=1}^{4} f_i(x)=2\pi-2$

(b) $\sum_{i=1}^{4}(-1)^i f_i(x)=0$

(c) $\sum_{i=1}^{4}(-1)^{\left[\frac{i}{3}\right]} f_i(x)=0$ (where [.] is greater integer function)

(d) $f_1(x) > f_2(x) \Rightarrow \cos 2x > 0$

11. If $f(x) = ax^2 + bx + c = 0$ have real roots and its coefficients are odd positive integers then

(a) $f(x) = 0$ always have irrational roots

(b) $\left| f\left(\frac{p}{q}\right) \right| \geq \frac{1}{q^2}$ where $p, q \in I$

(c) if $a.c = 1$, then equation must have exactly one root α such that $[\alpha] = -1$, where [.] is greatest integer function

(d) equation have rational roots

SECTION – III
Reasoning Type

This section contains 5 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

12. **Statement-1 :** $f(x) = \frac{\ln x}{(e^{-x} + 1)} (x > 1)$ is an increasing function.

Statement-2 : For $f(x) = \frac{g(x)}{h(x)}$, if $g(x)$ is an increasing function and $h(x)$ is a decreasing function, $g(x)$ and $h(x) > 0$, then $f(x)$ is an increasing function.

(a) A (b) B (c) C (d) D

13. **Statement-1 :** The minimum length of tangent drawn from point P on $3x+4y-20=0$ to the circle $x^2+y^2=1$ is $\sqrt{15}$.

Statement-2 : If tangent from point P touches the circle at T and secant passing through P intersect it at Q and R, then $PT^2 = PQ.PR$.

(a) A (b) B (c) C (d) D

14. **Statement-1 :** Let $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ and A_i is cofactor corresponding to a_i, then $\sum_{i=1}^{3} a_{3n+i} A_{3m+i} = 0, \quad (n \neq m \text{ and } n, m \in \{0, 1, 2\})$

Statement-2 : If a determinant has two rows identical, then its value is zero.

(a) A (b) B (c) C (d) D

15. **Statement-1 :** $\tan\left(\frac{\pi}{4}\left(\frac{1+\sin^2 x}{1+\sin^2 y}\right)\right) + \tan\left(\frac{\pi}{4}\left(\frac{1+\cos^2 x}{1+\cos^2 y}\right)\right) > 1$ for $x, y \in \left(0, \frac{\pi}{2}\right)$.

Statement-2: If $f(x,y) = \left(\frac{1+\sin^2 x}{1+\sin^2 y} - 1\right)$, then $f(x,y).f\left(\frac{\pi}{2}-x, \frac{\pi}{2}-y\right) \leq 0 \ \forall \ x, y \in \left(0, \frac{\pi}{2}\right)$.

(a) A (b) B (c) C (d) D

16. Consider the sequence $\frac{{}^nC_0}{1.2}, \frac{{}^nC_1}{2.3}, \frac{{}^nC_2}{3.4}, ...$

Statement-1 : For $n = 12$, 6th term of sequence is greatest.

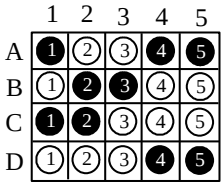
Statement-2: The general term of sequence is $T_r = \frac{{}^nC_{r-1}}{r.(r+1)}$ (for $r = 1, 2, ..., n$).

(a) A (b) B (c) C (d) D

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

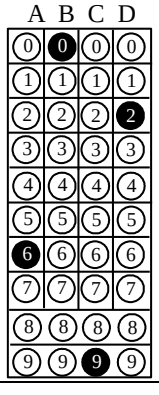
Column-I		Column-II
(A)	Let $S_n = \cos^4\theta + \cos^4\left(\theta + \frac{2\pi}{n}\right) + \cos^4\left(\theta + \frac{4\pi}{n}\right)$ + …. upto n terms, then S_n can be	**1.** 2
(B)	If in a ΔABC $\cos mA + \cos mB + \cos mC = 1 + 4\sin\frac{mA}{2}\sin\frac{mB}{2}\sin\frac{mC}{2}$, then m can be	**2.** 3
(C)	If $\cos^n x - \sin^n x = 1$ have 11 roots in $\left[0, \frac{23\pi}{2}\right)$, then n can be	**3.** 1
(D)	The values of m for which equation $(5m - m^2)^2 \sin^2 x - 10\sin x(5m - m^2) + 24 = 0$, has exactly three solutions in $[0, 2\pi]$ are	**4.** 6 **5.** 5

2.

Column-I	**Column-II**
(A) $\lim\limits_{n\to\infty}\dfrac{\left(\sum\limits_{x=1}^{n}x^4\right)\left(\sum\limits_{x=1}^{n}x^5\right)}{\left(\sum\limits_{x=1}^{n}x^t\right)\left(\sum\limits_{x=1}^{n}x^{9-t}\right)}=\dfrac{4}{5}$, then t can be	**1.** 6
(B) If m be the slope of tangent to the curve $x^y=y^x$ at (e, e), then $(3-m)$ can be	**2.** 2
(C) If $f:R-\{0\}\to R,\ f(x)f(y)=f(xy)+3\left(\dfrac{1}{x}+\dfrac{1}{y}\right)$, then $2\left\lvert f\left(-\dfrac{1}{2}\right)\right\rvert$ can be	**3.** 3
(D) $\lim\limits_{n\to\infty}\left\{(\sqrt{2}+1)^{n!+K}\right\}=1$, then K can be (where $\{.\}$ is a fractional part function)	**4.** 4 **5.** 7

SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



---

1. If the line $3x+4y=12$ intersects the hyperbola at P & P' and its asymptotes at Q and Q' then find the value of $\frac{PQ}{P'Q'}+\frac{PQ'}{P'Q}$.

Space for work

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2. Three students appear in three examinations. If the number of possible ways in which neither any student fails in all examinations nor there is an examination, which is not cleared by any student is n, then find $\frac{n}{53}$.

Space for work

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3. Let $f(x) = 30 - 2x - x^3$, then find the number of positive integral values of x which satisfies $f(f(f(x))) > f(f(-x))$.

Space for work

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4. Let $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$, if $\det.(A+A^2+A^3+A^4+\ldots+A^n)=64$, then find the value of n.

Space for work

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PAPER-I
PHYSICS
SECTION – I
Single Correct Choice Type

1. $(B_1)_x = 0,\ (B_1)_y = \frac{\mu_0 I}{4\pi d}\left(-\hat{j}\right)$

$(B_2)_x = \frac{\mu_0 I}{8\pi d}\sin\theta\hat{i},\ (B_2)_y = \frac{\mu_0 I}{8\pi d}\cos\theta\left(-\hat{j}\right)$

$(B_3)_x = \frac{\mu_0 I}{8\pi d}\cos\theta\left(-\hat{i}\right),\ (B_3)_y = \frac{\mu_0 I}{8\pi d}\sin\theta\left(-\hat{j}\right)$

$B_{net} = \frac{\mu_0 I}{8\pi d}\left[\sin\theta - \cos\theta\right]\hat{i} - \frac{\mu_0 I}{8\pi d}\left(2 + \cos\theta + \sin\theta\right)\hat{j}$

$B_{net} = \frac{\mu_0 I}{8\pi d}\sqrt{6 + 4\sin\theta + 4\cos\theta}$

$f(\theta) = 6 + 4\sin\theta + 4\cos\theta$

$f'(\theta) = 4(\cos\theta - \sin\theta) = 0$

$\tan\theta = 1$ for $0^0 < \theta < \frac{\pi}{2}$

$\theta = \frac{\pi}{4}$

$\therefore$ **(a)**

2. Angular momentum of the body about any point on the body

$\vec{L}_P = I_P\vec{\omega} = \left(\frac{MR^2}{2} + \frac{MR^2}{4}\right)\omega = \frac{3MRv_0}{4}$

$\therefore$ **(d)**

3. At equilibrium $2N\cos 30^0 = mg$

$\Rightarrow \quad N = \frac{mg}{\sqrt{3}}$

$\frac{2N}{2} = \frac{Kl}{2}$

$\frac{K}{m} = \frac{2g}{\sqrt{3}l}$

Applying energy concepts $-mgx + \frac{1}{2}K\left(\frac{x}{\sqrt{3}}\right)^2 + \frac{1}{2}mv^2 = E$

Differentiating we get $\frac{dE}{dt} = 0$

$$-mgv+\frac{Kx}{3}v+mv\frac{dv}{dt}=0$$

So $a=\frac{-2g}{3\sqrt{3}l}x+mg$

$$T=2\pi\sqrt{\frac{3\sqrt{3}l}{2g}}$$

$\therefore$ **(b)**

4. $(F_b-F_A)x=\frac{Mg(y-x)}{2}$

where $F_b=m_bg=\rho_{al}Vg$ (force of gravity = force of Buoyancy)

and $F_A=\rho_w g\frac{V}{2}$ and $V=\frac{4}{3}\pi r^3$

$\therefore$ $\frac{y}{x}=1.5$

$\therefore$ **(a)**

5. $T=\left(\frac{T_2-T_1}{l}\right)x+T_1$ if $(T_2>T_1)$

$v=k\sqrt{T}$

or $\frac{dx}{dt}=k\sqrt{T}$

$$\int_0^l \frac{dx}{\sqrt{\left(\frac{T_2-T_1}{l}\right)x+T_1}}=\int_0^t kdt$$

$\therefore$ **(a)**

6. Given that $P=2e^{2V}$

$$\frac{nRT}{V}=2e^{2V}$$

$$T=\frac{2V}{nR}e^{2V}$$

By differentiating $\frac{dT}{dV}=\frac{2}{nR}\left[2Ve^{2V}+e^{2V}\right]$

$$C=C_V+\frac{PdV}{ndT}=C_V+\frac{R}{1+2V}$$

$\therefore$ **(a)**

7. At steady state of the capacitor

$$I = \frac{\varepsilon}{3R}$$

also $V_B - V_C = 2\varepsilon$ and $V_0 - V_B = 2\varepsilon$

$\therefore$ $E_1 = \frac{1}{2}C_1(2\varepsilon)^2 = 2C_1\varepsilon^2$

$$E_2 = \frac{1}{2}C_2\left(2\varepsilon - \frac{\varepsilon}{3}\right)^2 = \frac{25C_2\varepsilon^2}{18} \text{ now } E_1 = E_2$$

$$2C_1\varepsilon^2 = \frac{25C_2\varepsilon^2}{18}$$

$\therefore$ $\frac{C_1}{C_2} = \frac{25}{36}$

$\therefore$ **(a)**

8. By distribution of charge, all the capacitor of 1μF in between the above and lower branch is removed, then the equivalent capacitance of upper branch is

$$\frac{1}{C_{eq}} = \frac{1}{1} + \frac{1}{3} + \frac{1}{9} + ...$$

$$C_{eq} = \frac{1 - \frac{1}{3}}{1} = \frac{2}{3}$$

$\therefore$ $C_{net} = 2 \times \frac{2}{3} = \frac{4}{3}\mu F$

$\therefore$ **(b)**

SECTION – II
Multiple Correct Choice Type

9. $\frac{\Delta\theta}{\Delta x} = \frac{H}{AK}$

Heat flow is $\propto \frac{1}{AK}$

$\therefore$ **(a, c)**

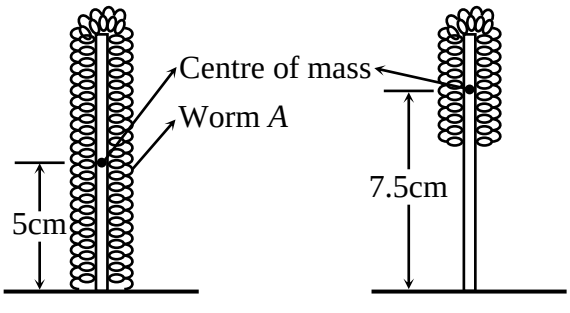
10. On worm *A* work done by gravity

$W_1 = -\Delta U_1$

For worm *B*, work done by worm

$W_2 = \Delta U_2$

$\therefore$ **(a, c)**



11. Visible range is from 3800 Å to 7800 Å and the wavelength 6000 Å corresponds to yellow colour so blue and violet colour can cause photoelectric emission, whatever be the intensity.

$\therefore$ **(b, c, d)**

12. For $\lambda = 1,\ dU + W = 0 \Rightarrow Q = 0$

For $\lambda = 0,\ dU = 0 \quad \Rightarrow$ T = constant

For $\lambda = -\frac{5}{2},\quad dU = \frac{5}{2}W$

$Q = dU + W = \frac{7}{2}W$

$\frac{Q}{dU} = \frac{7}{5} \quad \Rightarrow \quad Q = \frac{7}{5} \times n\frac{5}{2}RdT$

$Q = nC_p dT \quad \Rightarrow$ Pressure is constant

For $\lambda = \frac{5}{2}$,

$\Rightarrow \quad W = -\frac{2}{5}dU = \frac{-2}{5}(nCvdT)$

$W = -nRdT = \frac{P_2V_2 - P_1V_1}{-1}$

If PV^N = constant

$\Rightarrow \quad W = \int_{v_1}^{v_2} P\,dV = \frac{P_2V_2 - P_1V_1}{1-N}$

Solving $1 - N = -1$

$\Rightarrow \quad N = 2$

$\therefore$ **(a, c, d)**

SECTION–III
Comprehension Type
Passage-I

13. $2mv_0 = (2m + M)v' \Rightarrow v' = \frac{2v_0}{3}$

$\therefore$ **(d)**

14. $\frac{1}{2}(2m)v_0^2 + \frac{q^2}{4\pi \in_0 r} = \frac{1}{2}(3m)\left(\frac{2v_0}{3}\right)^2 + U$

$\therefore$ **(b)**

15. $2mv_0 = 2mx + m\left(\frac{v_0}{2}\right)\ ;\ \frac{3mv_0}{2} = 2mx\ ;\ x = \frac{3v_0}{4}$

$\frac{1}{2}(2m)v_0^2 + \frac{q^2}{4\pi \in_0 r} = \frac{q^2}{4\pi \in_0 x_1} + \frac{1}{2}(2m)\left(\frac{3v_0}{4}\right)^2 + \frac{1}{2}m\left(\frac{v_0}{2}\right)^2$

$mv_0^2 - \frac{11}{16}mv_0^2 + \frac{q^2}{4\pi \in_0 r} = \frac{q^2}{4\pi \in_0 x_1}$

$$\frac{5mv_0^2}{16}+\frac{q^2}{4\pi\in_0 r}=\frac{q^2}{4\pi\in_0 x_1}$$

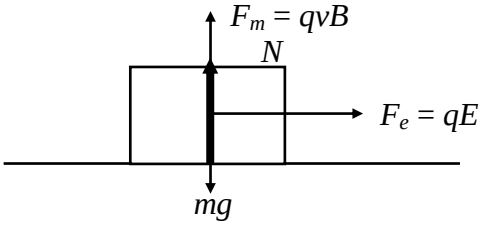
$$x_1=\frac{q^2}{4\pi\in_0\left(\frac{q^2}{4\pi\in_0 r}+\frac{5mv_0{}^2}{16}\right)}.$$

$\therefore$ **(a)**

Passage-II

Initially the charged block will experience gravitational and electric forces, but no magnetic force being at rest. Electric force will accelerate the block from rest.

At any instant when it moves



$F_m = qVB \left[\because \vec{V} \perp \vec{B}\right]$

$N + qvB = mg$

$N = mg - qvB = mg - q\left\{\frac{qE}{m}t\right\}B$ $\left[\because a = \frac{F_e}{m} \text{ and } v = at\right]$

$N = mg - \frac{q^2EBt}{m}$ … (i)

16. The block will lose contact when $N = 0$

From equation (i) $t = \frac{m^2 g}{q^2 EB}$

Putting values $t = 100$ sec

$\therefore$ **(c)**

17. Distance travelled upto $t = 100$ sec

$S = \frac{1}{2}\left\{\frac{qE}{m}\right\}t^2 = 5 \times 10^5$ m

$\therefore$ **(d)**

18. From equation (i) magnitude of time rate of change of N

$\left|\frac{dN}{dt}\right| = \frac{q^2EB}{m} = 10^{-2}$ N/sec

$\therefore$ **(a)**

SECTION–IV
Matrix–Match Type

1. **(A)** $\vec{v} = 5t\,\hat{i} + 4t\,\hat{j}$

$\vec{a} = \dfrac{d\vec{v}}{dt} = 5\hat{i} + 4\,\hat{j}$

So acceleration is constant

Also $\vec{v}\,(\text{at } t = 0) = 0$

So particle moves in a straight line.

(B) $\vec{v} = -5t\,\hat{i} + 5\,\hat{j}$

$\dfrac{d\vec{v}}{dt} = -5\hat{i}$ acceleration is constant and along $-x$ axis.

Now initial velocity is $5\,\hat{j}$

So particle moves in parabolic path.

(C) $y = 2x - 5x^2$

This is the equation of a parabola.

Hence path of the particle is parabolic. A particle which experience constant acceleration will follow parabolic path if projected in a direction other then the direction of acceleration.

(D) $x = 6\sin\dfrac{2\pi t}{5}$, $y = 6\cos\dfrac{2\pi t}{5}$

$x^2 + y^2 = 36$

Path is circular with constant speed.

$\therefore$ **(A) – (1, 3, 4), (B) – (2, 3, 4), (C) – (2, 3, 4), (D) – (4, 5)**

2. For He^+

$$E_n = \frac{-13.6\,Z^2}{n^2}\text{eV},\ E_1 = \frac{-13.6\times Z^2}{1^2} = -54.4\,\text{eV}$$

As reference has been shifted

$$E_n = \frac{-13.6\,Z^2}{n^2}\text{eV} + 54.4\,\text{eV}$$

For $n = 2,\ E_2 = \dfrac{-13.6\times Z^2}{2^2} + 54.4\,\text{eV} = 40.8\,\text{eV}$

Ionization energy from ground state is 54.4 eV

To excite an electron from $n = 1$ to $n = 2$

$\Delta E = -13.6\text{ eV} - (-54.4\text{ eV}) = 40.8\text{ eV}$

$$U_n = \frac{-27.2\,Z^2}{n^2} + 54.4\,\text{eV} = -54.4\,\text{eV}$$

$\therefore$ **(A) – (2, 3, 5), (B) – (1, 4), (C) – (2, 4, 5), (D) – (1, 3)**

CHEMISTRY

SECTION – I
Straight Objective Type

1. **(a)** Precipitation of NaCl will take place after solution become saturated.

Molarity of sample of sea water, $M = \frac{W_{NaCl}}{M_{NaCl}} \times \frac{1000}{V_{in\ mL}} = \frac{2.8}{58.5} \times \frac{1000}{100/1.03} = 0.493\ M$

$M_1V_1 = M_2V_2$

$0.493 \times 4 = 5.45 \times V_2$

$V_2 = 0.361\ L$

Water to be evaporated would be = (4 – 0.361)L = 3.639 L ≃ **3.64 L**.

2. **(c)** The cell is $Pt \mid H_2 \mid H^+(1\ M) \parallel Ag_2SO_4$ (saturated) $\mid Ag(s)$

At anode: $H_2 \longrightarrow 2H^+ + 2e^-$

At cathode: $2Ag^+ + 2e \longrightarrow Ag$

$H_2 + 2Ag^+(aq) \longrightarrow 2H^+(aq) + Ag(s)$

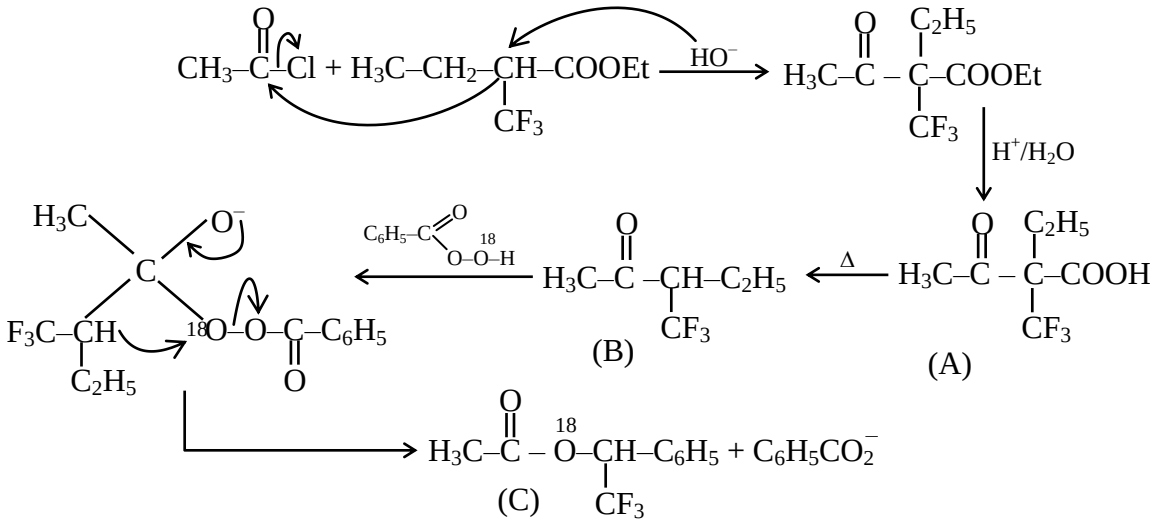
$E_{cell} = E^o_{cell} - \frac{0.059}{2}\log\frac{[H^+]^2}{[Ag^+]^2} \times \frac{1}{p_{H_2}}$

$0.74 - 0.799 = -0.059\ \log\frac{1}{[Ag^+]\times 1}$; $1 = \log\frac{1}{[Ag^+]}$; $[Ag^+] = 10^{-1}\ M$

$Ag_2SO_4(s) \rightleftharpoons 2Ag^+(aq) + SO_4^{2-}(aq)$

$[Ag^+] = 10^{-1}$ and $[SO_4^{2-}] = \left[\frac{10^{-1}}{2}\right]$; $K_{sp} = [Ag^+]^2\ [SO_4^{2-}] = [10^{-1}]^2 \left[\frac{10^{-1}}{2}\right] = \mathbf{5 \times 10^{-4}}$.

3. **(c)**



4. **(c)** Total number of ^{40}K atoms in 80 kg human body

$$= \frac{1.2\times10^{-2}}{100}\times\frac{3.5\times10^{-1}}{100}\times80\times10^{3}\times\frac{6.02\times10^{23}}{40}$$

$= 5.06 \times 10^{20}$ atoms

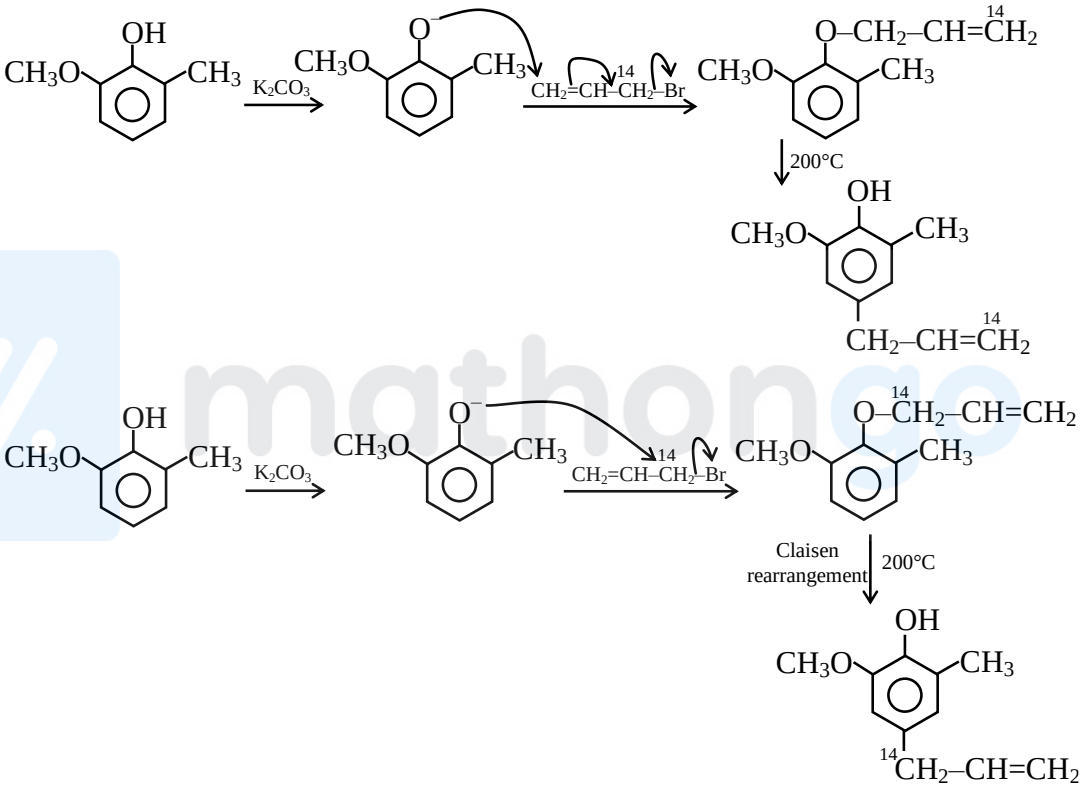
$$\lambda = \frac{0.693}{6.9\times10^{14}} = 1.004 \times 10^{-15} \simeq 10^{-15}.$$

$A = N\lambda$

$= 5.06 \times 10^{20} \times 10^{-15} =$ **5.06×10^{5} dpm**.

SECTION – II
Multiple Correct Choice Type

5. **(a, b)**



6. **(b, c)**

$$Na_2S + Na_2SO_3 + I_2 \longrightarrow \underset{(A)}{Na_2S_2O_3} + 2NaI$$
(B)

$$Na_2S_2O_3 + AgNO_3 \longrightarrow \underset{\text{(white ppt.)}}{Ag_2S_2O_3} + 2NaNO_3$$

$$Ag_2S_2O_3 + H_2O \longrightarrow \underset{\text{(black ppt.)}}{\underset{(C)}{Ag_2S}} + H_2SO_4$$

7. **(b, d)** Vapour pressure of solution is inversely proportional to its boiling point. ∴, boiling point of (A) would be lower than (B), as its vapour pressure is higher than (B).

If (A) and (B) solution have shown positive deviation than it would form minimum boiling point azeotrope.

8. **(a, b, c)**

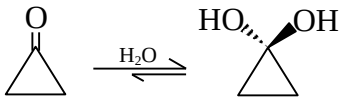
$NH_4Cl + NaNO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

$3CaOCl_2 + 2NH_3 \xrightarrow{\Delta} N_2 + 3CaCl_2 + 3H_2O$

$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$

$CaCN_2 + 4H_2O \xrightarrow{\Delta} Ca(OH)_2 + CO_2 + 2NH_3$

SECTION – III
Reasoning Type

9. **(b)** 

Driving force of reaction is provided by the measure of relief in bond strain on going from carbonyl compound (where C–C–C bond angle is 60° compared to normal sp^2 value of 120°) to hydrate (C–C–C bond angle is 60° compared to normal sp^3 value of 109°).

10. **(a)** $Hg_2(NO_3)_2 \longrightarrow Hg_2^{2+} + 2NO_3^-$

11. **(b)** At 25°C, the ionic product of H_2O and D_2O are 10^{-14} and 3×10^{-15} respectively justifying that dissociation of H_2O is much more than that of D_2O giving more H^+ ions than D^+ ons.

12. **(b)** Due to low K_{sp} of $BaSO_4$ in water, it is nearly insoluble in water. So, it is not absorbed by body and we do not get any toxicity.

SECTION–IV
Comprehension Type

Passage-I

13. **(a)** Molecular weight of (A) = $\frac{0.369 \times 22400}{67.2} = 123$

Let bromo derivative be R–Br.

Molecular weight of R = 123 – 80 = 43

$C_nH_{2n+1} = 43$; $n = 3$

$\therefore$ Molecular formula = **C_3H_7Br**.

14. **(d)** (A) is C_3H_7Br which may exist as $CH_3CH_2CH_2Br$ or $CH_3CH(Br)CH_3$.

$CH_3CH_2CH_2Br$ (A) $\xrightarrow{NaOH}$ $CH_3CH_2CH_2OH$ (B)

(B) $\xrightarrow[250°C]{Alumina}$ $CH_3CH_2CH_2OCH_2CH_2CH_3$ (C)

(B) $\xrightarrow[350°C]{Alumina}$ $CH_3CH=CH_2$ (D)

$$\underset{(D)}{CH_3CH=CH_2} \xrightarrow{HBr} \underset{\text{(Isomer of A)}}{CH_3CH(Br)CH_3} \xrightarrow{\text{Alco. KCN}} CH_3-\underset{\underset{CN}{|}}{CH}-CH_3 \xrightarrow[-NH_3]{2H_2O/H^+} \underset{\underset{COOH}{|}}{CH_3-CH-CH_3}\ (E)$$

$$\underset{\underset{COOH}{|}}{CH_3-CH-CH_3}\ (E) \xrightarrow{Br_2/\text{Red P}} \underset{\underset{COOH}{|}}{CH_3-C(Br)-CH_3}\ \text{(Optically inactive)}$$

If (A) would be $CH_3–CH(Br)–CH_3$ then isomer of (A) is $CH_3–CH_2–CH_2–Br$ and (E) is $CH_3–CH_2–CH_2–COOH$ which would give optical active compound $CH_3–CH_2–CH(Br)–COOH$ on its HVZ reaction.

15. **(d)** $\underset{}{CH_3CH_2CH_2OCH_2CH_2CH_3} \xrightarrow{HBr} \underset{(A)}{CH_3CH_2CH_2Br} + \underset{(B)}{CH_3CH_3CH_2OH}$

Passage-II

16. **(c)** $P = \dfrac{829 \text{ mm of Hg}}{760 \text{ mm of Hg}} = 1.09 \text{ atm}$

Mole of N_2 required $= \dfrac{PV}{RT} = \dfrac{(1.09 \text{ atm})\,(45.5\,L)}{(0.0821\,L \text{ atm } K^{-1} \text{ mol}^{-1})\,(295\,K)} = 2.05 \text{ mol of } N_2$

$2NaN_3(s) \longrightarrow 2Na + 3N_2(g)$

3 mole of N_2 are produced from = 2×65 g of NaN_3

2.05 mole of N_2 are produced from = $\dfrac{2\times65}{3}\times 2.05$ = **88.8 g NaN_3**.

17. **(a)** Applying $\dfrac{P_1V_1}{T_1} = \dfrac{P_2V_2}{T_2}$

$V_2 = V_1 \times \dfrac{P_1}{P_2}\times\dfrac{T_2}{T_1}$

$= 4.19 \times 10^3 \times \dfrac{754}{76}\times\dfrac{240}{295.5}$ = **3.38×10^4 L**.

18. **(d)**

SECTION–IV
Matrix–Match Type

1. (A) – 1, 3, 4 ; (B) – 2, 3 ; (C) – 4, 5 ; (D) – 2, 3, 4

Sol of $Fe(OH)_3$ containing $FeCl_3$ will form positive colloid [i.e. $Fe(OH)_3.Fe^{3+}$] which can be formed by peptization and chemical method.

By hydrolysis: $FeCl_3 + 3H_2O \longrightarrow \underset{\text{(Collidal sol)}}{Fe(OH)_3} + 3HCl$

$Fe(OH)_3$ sol can be obtained by boiling solution of its chloride ($FeCl_3$).

A precipitate of silver iodide can be peptized by shaking with a dilute solution of KI and due to excess of I^- ion it will be negative colloid which may be represented as $AgI.I^-$.

Colloidal sol of gold metal can be obtained by Bredig Arc method and by reduction of very dilute solution of their salts with a suitable reducing agents e.g.

$$2AuCl_3 + 3SnCl_2 \longrightarrow \underset{\text{(Gold sol)}}{2Au} + 3SnCl_4$$

As_2S_3 in H_2S in negative collid due to excess of sulphuide ion, chemically obtained by double decomposition.

$$As_2O_3 + 3H_2S \longrightarrow \underset{\text{(Collidal sol)}}{As_2S_3} + 3H_2O$$

2. (A) – 1, 2 ; (B) – 1, 2, 3 ; (C) – 2, 4, 5 ; (D) – 2

State functions are those vairables whose value depends on initial and final state of system.

Extensive property are those whose value depends on amount of substance.

$dq = du + dw$

$TdS = d(H - PV) + PdV$; $TdS = dH - PdV - VdP + PdV$; $TdS + VdP = dH$

At constant P, $TdS = dH$

$$T = \left(\frac{dH}{dS}\right)_P$$

MATHEMATICS

SECTION – I

Single Correct Choice Type

1. (b) Total number of cases = $^{20}C_{10} \times 10! \times 10!$

Favorable cases = $^{10}C_5 \times 10! \times 10!$

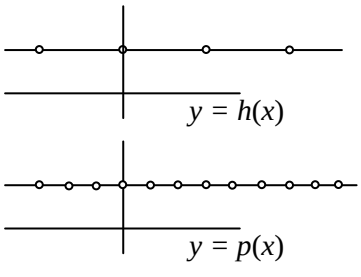
Hence required probability = $\frac{^{10}C_5}{^{20}C_{10}}$

Consider that after each draw, the person arrange the balls in the order.

2. (b) $h(x) = \frac{\{x\}}{\{x\}}$

$p(x) = \frac{g(x)}{g(x)}$

Obviously the combined period will be 1/4 if $g(x)$ becomes zero at equal interval of 1/4 and as well as at the integers.



3. (b) $f'\left(\frac{x}{y}\right).f\left(\frac{y}{x}\right) = \frac{x^2+y^2}{xy}$ …(i)

Interchanging x, y we get

$$f'\left(\frac{x}{y}\right).f\left(\frac{y}{x}\right) = f'\left(\frac{y}{x}\right)f\left(\frac{x}{y}\right) \Rightarrow \ln f\left(\frac{x}{y}\right) = \ln f\left(\frac{y}{x}\right) + C$$

Putting $x = y$, we get $C = 0 \Rightarrow f\left(\frac{x}{y}\right) = f\left(\frac{y}{x}\right)$ …(ii)

Putting $y = 1,$ in equation (i) and (ii), we get

$$\Rightarrow f'(x)f\left(\frac{1}{x}\right) = \frac{x^2+1}{x} \text{ and } f(x) = f\left(\frac{1}{x}\right)$$

$$\Rightarrow f'(x)f(x) = x + \frac{1}{x} \Rightarrow (f(x))^2 = x^2 + 2\ln x$$

4. (b) $(x^2+2x+4)^n = \sum_{r=0}^{2n} a_r x^r$

Let $x = 2t \Rightarrow 4^n(1+t+t^2)^n = \sum_{r=0}^{2n} 2^r a_r t^r$

Let $x = \frac{2}{t} \Rightarrow 4^n(1+t+t^2)^n = \sum_{r=0}^{2n} 2^r a_r t^{2n-r}$

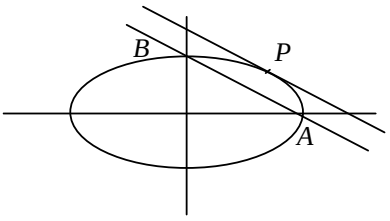
On comparing coefficient of t_r

$$\Rightarrow 2^r a_r = 2^{2n-r} a_{2n-r} \Rightarrow \sum_{r=n}^{2n} \frac{a_{2n-r}}{a_r} = \sum_{r=n}^{2n} \frac{1}{4^{n-r}} = \frac{4^{n+1}-1}{3}$$

5. **(c)** Let perpendicular distance of P from the line be h

$\frac{1}{2}\times h\times 5 = 6(\sqrt{2}-1)$ (as $\Delta PAB = 6(\sqrt{2}-1)$)

$\Rightarrow \quad h = \frac{12(\sqrt{2}-1)}{5}$



Now distance of tangent parallel to AB i.e. $4y+3x=12\sqrt{2}$, from line AB is $\frac{12(\sqrt{2}-1)}{5}$.

There are just three such points.

6. **(d)** $z = a+ib$

$\Rightarrow \quad |a+ib|^{2010} = |a-ib|$

$\Rightarrow \quad |z|^{2010} = |z| \quad \Rightarrow \quad |z| = 0 \quad \Rightarrow \quad (a, b) \equiv (0, 0)$

or $\quad |z| = 1 \quad \Rightarrow \quad z^{2010} = \frac{1}{z} \quad \Rightarrow \quad z^{2011} = 1$

Hence z has 2012 roots.

SECTION – II
Multiple Correct Choice Type

7. **(a), (c)**

$$f\left(g(n, h(x))\right) = \left|\sin\pi\left([n]+[n]^2\right)^{\frac{x}{\log\left(\frac{e^x+e^{2x}+e^{3x}}{3}\right)}}\right|$$

$$\lim_{x\to 0} f(g(n, h(x)) = \left|\sin\pi([n]+[n]^2)^{\frac{1}{2}}\right|$$

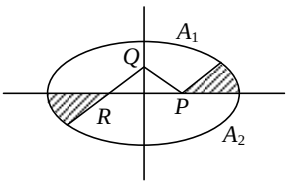
$$\Rightarrow \quad \lim_{n\to\infty}\sin\left(\pi[n]\left(1+\frac{1}{[n]}\right)^{1/2}\right) = \lim_{n\to\infty}\left|\sin\left(\pi[n]+\frac{\pi}{2}-\frac{1}{[n]}\times\frac{\pi}{8}\ldots\right)\right| = 1$$

$$\lim_{n\to\infty}\lim_{x\to 0} g(n,\ h(x)) = \lim_{n\to\infty}\sin\left(\pi[n]+\frac{\pi}{2}-\frac{1}{8[n]}+\ldots\right)$$

does not exists as value can be 1 or –1.

8. **(a), (c)**

Area of ellipse $= \pi\times\frac{8}{3\sqrt{\pi}}\times\frac{5}{2\sqrt{\pi}} = \frac{20}{3} = A_e$



Both shaded areas are equal, then

$A_1 = \frac{1}{2}A_e - \Delta PQR = \frac{10}{3} - 1$

$A_2 = \frac{1}{2}A_e + \Delta PQR = \frac{10}{3} + 1$

9. (a), (b), (c)

If n is even

$$f(n) = 2\left(1+2+\ldots\frac{n-2}{2}\right)+\frac{n}{2} = \frac{n^2}{4}$$

If n is odd then $f(n) = f(n-1)+\frac{n-1}{2} = \frac{(n-1)^2}{4}+\frac{n-1}{2} = \frac{n^2-1}{4}$

$$f(n+m)-f(n-m) = \frac{(n+m)^2-(n-m)^2}{4} \text{ if both even}$$

$$= \frac{(n+m)^2-1-(n-m)^2+1}{4} \text{ it both odd}$$

$$= nm$$

10. (a), (b), (c) & (d)

$$f_1(x) = \frac{\pi}{2}-\sin^2 x,\quad f_2(x) = \frac{\pi}{2}-\cos^2 x$$

$$f_3(x) = \frac{\pi}{2}-\cos^2 x,\quad f_4(x) = \frac{\pi}{2}-\sin^2 x$$

11. (a), (b) & (c)

If a, b, c are odd integer then D can not be perfect square,

$\Rightarrow \quad |f(\lambda)| > 0$ where $\lambda \in Q$

$\Rightarrow \quad |a p^2+b pq+c q^2| \geq 1 \quad \Rightarrow \quad \left|f\left(\frac{p}{q}\right)\right| \geq \frac{1}{q^2}$

If $a.c = 1 \quad \Rightarrow \quad a = c = 1$ and $b \geq 3$ as

$c.(a-b+c) < 0 \quad \Rightarrow \quad \alpha \in (-1, 0)$

SECTION–III
Reasoning Type

12. (a) $f'(x) = \frac{h(x)g'(x)-h'(x)g(x)}{(h(x))^2} > 0$

(As $h(x), g(x) > 0, g'(x) > 0$ and $h'(x) < 0$)

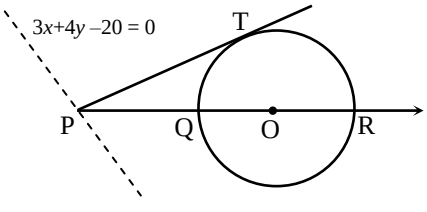
13. (a) Let $OP = x$

$PT^2 = PQ.PR = (OP-1):(OP+1) = (OP)^2-1$

For PT minimum OP should be minimum.

Minimum distance of OP from (0, 0) to $3x+4y-20=0$ is 4

$\Rightarrow \quad PT^2 = 16-1 = 15 \quad \Rightarrow \quad PT = \sqrt{15}$



14. (a) The value of cofactors of elements of particular row remain unchanged if we change a particular row with any other row, which will make two row identical and hence value of change determinant will become zero.

15. (a) $f(x,\ y)\, f\left(\frac{\pi}{2}-x, \frac{\pi}{2}-y\right)$

$$\Rightarrow \quad \left(\frac{1+\sin^2 x}{1+\sin^2 y}-1\right)\left(\frac{1+\cos^2 x}{1+\cos^2 y}-1\right)=\frac{(\sin^2 x-\sin^2 y)(\cos^2 x-\cos^2 y)}{(1+\sin^2 y)(1+\cos^2 y)} \le 0$$

$$(\text{as } (\sin x-\sin y)(\cos x-\cos y)\le 0,\ \ \forall x,\ y\in(0,\pi/2))$$

$\Rightarrow$ Exactly one of $(f(x,\ y)+1), \left(f\left(\frac{\pi}{2}-x, \frac{\pi}{2}-y\right)+1\right)$ is greater than 1.

$\Rightarrow$ Statement 1 is true.

16. (b) Let $T_{r+1} \ge T_r$

$$\Rightarrow \quad \frac{{}^nC_r}{(r+1)(r+2)} \ge \frac{{}^nC_{r-1}}{r.(r+1)}$$

$$r \le \frac{n-1}{2} \ \Rightarrow \ r \le 5.5$$

S0 6th term of sequence greatest.

SECTION–IV
Matrix–Match Type

1. (A) – 2, 4; (B) – 3, 5; (C) – 2, 3, 5; (D) – 1, 2, 4

(A) $S_n = \sum_{r=1}^{n} \cos^4\left(\theta+\frac{2(r-1)\pi}{n}\right)$

$$=\frac{1}{4}\sum_{r=1}^{n}\left(1+\cos\left(2\theta+\frac{4(r-1)\pi}{n}\right)\right)^2$$

$$=\frac{1}{4}\sum_{r=1}^{n}\left(1+\cos^2\left(2\theta+\frac{4(r-1)\pi}{n}\right)+2\cos\left(2\theta+\frac{4(r-1)\pi}{n}\right)\right)$$

$$=\frac{1}{4}\sum_{r=1}^{n}\left(1+\frac{1}{2}\left(1+\cos\left(4\theta+\frac{8(r-1)\pi}{n}\right)\right)\right)$$

$$=\frac{1}{4}\times\frac{3}{2}n=\frac{3}{8}n$$

(B) $\cos mA+\cos mB+\cos mC=2\cos m\left(\frac{A+B}{2}\right)\cos m\left(\frac{A-B}{2}\right)+1-2\sin^2\frac{mC}{2}$

$=1+2\sin\frac{mC}{2}\left(\cos m\left(\frac{A-B}{2}\right)-\sin\frac{mC}{2}\right)$ for $m=(4n+1)$

$=1+2\sin\frac{mC}{2}\left(\cos m\left(\frac{A-B}{2}\right)-\cos m\left(\frac{A+B}{2}\right)\right)$

$=1+4\sin\frac{mA}{2}\sin\frac{mB}{2}\sin\frac{mC}{2}$

(C) $\cos^n x=1+\sin^n x$, for n even $x=m\pi$,

For n odd $x=2m\pi$ or $x=\left(2p\pi+\frac{3\pi}{2}\right)$

(as $\cos^n x+\sin^n x\le\sin^2 x+\cos^2 x=1$ if $n>2$)

(D) $\sin x=\frac{6}{5m-m^2}$ or $\frac{4}{5m-m^2}$, but for exactly three roots $\sin x=\pm1$

$\Rightarrow \quad \frac{6}{5m-m^2}=\pm1 \quad \Rightarrow \quad m=2,\ 3,\ -1,\ 6$

2. (A) – 2, 5; (B) – 2, 4; (C) – 2, 3; (D) – 1, 2, 4

(A) $\lim\limits_{n\to\infty}\frac{\left(\sum\limits_{x=1}^{n}x^4\right)\left(\sum\limits_{x=1}^{n}x^5\right)}{\left(\sum\limits_{x=1}^{n}x^t\right)\left(\sum\limits_{x=1}^{n}x^{9-t}\right)}=\frac{\int_0^1 y^4dy\int_0^1 y^5dy}{\int_0^1 y^tdy\int_0^1 y^{9-t}dy}=\frac{(t+1)(10-t)}{30}=\frac{4}{5}$

$\Rightarrow \quad t^2-9t+14=0, \qquad (t-2)(t-7)=0, \ \ t=2,7$

(B) $x^y=y^x \quad \Rightarrow \quad y\ln x=x\ln y$

Differentiating with respect to to x we get $y'=\frac{(x\ln y-y)}{(y\ln x-x)}=\frac{y^2(\ln x-1)}{x^2(\ln y-1)}$

$y'=\lim\limits_{\substack{x\to e\\y\to e}}\frac{y^2(\ln x-1)}{x^2(\ln y-1)}=\lim\limits_{\substack{x\to e\\y\to e}}\frac{y}{x\ y'}$

$\Rightarrow \quad y'=\pm1$

$\Rightarrow$ So $m=\pm1$ and $3-m$ is equal to 2 or 4

(C) Put $x = y = 1$

$\Rightarrow \quad (f(1))^2 = f(1) + 6 \quad \Rightarrow \quad f(1) = 3, -2$

Put $y = 1$, we get $f(x) = \frac{3}{2}\left(\frac{1+x}{x}\right)$ or $f(x) = -\left(\frac{1+x}{x}\right)$

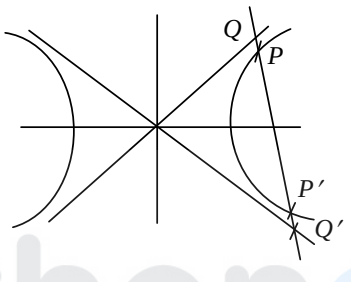
(D) If m is even $(\sqrt{2}+1)^m + (\sqrt{2}-1)^m = 2I$

$\Rightarrow \quad \lim\limits_{n\to\infty}\{(\sqrt{2}+1)^{n!+k}\} = \lim\limits_{n\to\infty}\{2I - (\sqrt{2}-1)^{n!+k}\} = \lim\limits_{n\to\infty}\{-(\sqrt{2}-1)^{n!+k}\} = 1$

$\Rightarrow \quad n!+k$ must be even integer $\Rightarrow$ k is even.

SECTION–V
Integer Answers Type

1. For any line in case of hyperbola we know that $PQ = P'Q'$



$$\frac{PQ}{P'Q'} + \frac{PQ'}{P'Q} = 1 + \frac{PP' + P'Q'}{PP' + PQ} = 2$$

Ans. 2

2. The number of ways when no student failed in all the examination $= (2^3 - 1)^3$

The number of ways when out of above cases atleast one examination was not cleared by any students $= {}^3C_1(3)^3$

The number of ways when out of above cases atleast any two examination were not cleared by any student $= {}^3C_2(1)^3$

So required cases $= (2^3 - 1)^3 - {}^3C_1 3^3 + {}^3C_2 = 265 = n$

Ans. 5

3. $f(x) = 30 - 2x - x^3$

$f'(x) = -2 - 3x^2 < 0 \quad \Rightarrow \quad f(x)$ is decreasing function

Hence $f(f(f(x))) > f(f(-x)) \quad \Rightarrow \quad f(f(x)) < f(-x)$

$\Rightarrow \quad f(x) > -x$

$\Rightarrow \quad 30 - 2x - x^3 > -x \Rightarrow x^3 + x - 30 < 0 \Rightarrow (x-3)(x^2+3x+10) < 0$

$\Rightarrow \quad x < 3$

Ans. 2

4. $\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}^k = \left(\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}+\begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix}\right)^k = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}+k\begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix} \quad \left(\text{as } \begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix}^2 = \text{O}\right)$

$$= \begin{bmatrix}2k+1 & -4k\\ k & 1-2k\end{bmatrix}$$

$$\Rightarrow \quad \sum_{k=1}^{n} A^k = \begin{bmatrix}n+n(n+1) & -2n(n+1)\\ \dfrac{n(n+1)}{2} & n-n(n+1)\end{bmatrix} \Rightarrow \quad \det.\left(\sum_{k=1}^{n} A^k\right) = n^2 = 64$$

$$\Rightarrow \quad n = 8$$

Ans. 8